

✓ Experiment 11

Aim: Understand basic NLP concepts and apply text preprocessing techniques.

```
import pandas as pd
import nltk

from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
```

```
nltk.download('punkt')
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
True
```

```
columns = [
    "Class",
    "Alcohol",
    "Malic_Acid",
    "Ash",
    "Alcalinity_of_Ash",
    "Magnesium",
    "Total_Phenols",
    "Flavanoids",
    "Nonflavanoid_Phenols",
    "Proanthocyanins",
    "Color_Intensity",
    "Hue",
    "OD280_OD315",
    "Proline"
]

df = pd.read_csv("wine.data", header=None, names=columns)
```

```
df['text'] = df.apply(
    lambda row: f"Wine with alcohol {row['Alcohol']} and malic acid {row['Malic_Acid']} and has ash {row['Ash']} and color intensity {row['Color_Intensity']}",
    axis=1
)

print(df['text'].head())
```

```
0    Wine with alcohol 14.23 and malic acid 1.71 ha...
1    Wine with alcohol 13.2 and malic acid 1.78 has...
2    Wine with alcohol 13.16 and malic acid 2.36 ha...
3    Wine with alcohol 14.37 and malic acid 1.95 ha...
```

```
4 Wine with alcohol 13.24 and malic acid 2.59 ha...
Name: text, dtype: object
```

NLP PREPROCESSING

Lowercasing

```
df['text_lower'] = df['text'].str.lower()
```

Tokenization

```
import nltk

nltk.download('punkt')
nltk.download('punkt_tab') # ← IMPORTANT (your error needs this)
nltk.download('stopwords')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data] Unzipping tokenizers/punkt_tab.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data] Package stopwords is already up-to-date!
True
```

```
df['tokens'] = df['text_lower'].apply(word_tokenize)
```

Stopword Removal

```
stop_words = set(stopwords.words('english'))

df['no_stopwords'] = df['tokens'].apply(
    lambda words: [w for w in words if w not in stop_words]
)
```

Stemming

```
stemmer = PorterStemmer()

df['stemmed'] = df['no_stopwords'].apply(
    lambda words: [stemmer.stem(w) for w in words]
)
```

DISPLAY RESULTS

```
print("\nOriginal Text:\n", df['text'].iloc[0])
print("\nLowercase:\n", df['text_lower'].iloc[0])
print("\nTokens:\n", df['tokens'].iloc[0])
```

```
print("\nAfter Stopword Removal:\n", df['no_stopwords'].iloc[0])  
print("\nAfter Stemming:\n", df['stemmed'].iloc[0])
```

Original Text:

Wine with alcohol 14.23 and malic acid 1.71 has ash 2.43 and color intens

Lowercase:

wine with alcohol 14.23 and malic acid 1.71 has ash 2.43 and color intens

Tokens:

['wine', 'with', 'alcohol', '14.23', 'and', 'malic', 'acid', '1.71', 'has

After Stopword Removal:

['wine', 'alcohol', '14.23', 'malic', 'acid', '1.71', 'ash', '2.43', 'co

After Stemming:

['wine', 'alcohol', '14.23', 'malic', 'acid', '1.71', 'ash', '2.43', 'co

Word Frequency

```
from collections import Counter
```

```
all_words = [word for row in df['no_stopwords'] for word in row]
```

```
word_freq = Counter(all_words)
```

```
print(word_freq.most_common(10))
```

```
[('wine', 178), ('alcohol', 178), ('malic', 178), ('acid', 178), ('ash', 1
```